

TONG BAO

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Employment

- **University of Potsdam** *Sep 2026 – present*
  Postdoc, Massive stars and their feedback in metal-poor galaxies Germany
- **INAF - Osservatorio Astronomico di Brera** *Sep 2024–Aug 2026*
  Postdoc, Galactic X-ray emission and feedback (FARE/HotMilk) Italy

Education

- **Nanjing University** *Sep 2019 – June 2024*
  Ph.D., Astrophysics Advisor: Prof. Zhiyuan Li

  Thesis: The Study of X-ray Periodic Variability in the Galactic Bulge and Globular Clusters

- **Nanjing University** *Sep 2015 – June 2019*
  B.S., Astronomy
  Thesis: Application of Neural Network Algorithm in X-ray Point Source Detection

Research experience and Collaborations

- **Member of the HotMilk group at INAF** *June 2024-now*
  Investigating the hot gaseous halo of the Milky Way.
- **Columbia University** *April 2023 , New York*
  Periodic X-ray sources and exotic binaries in dense environment Host: Prof. Charles J. Hailey & Dr. Kaya Mori
- **Columbia University** *June - Sep 2018, New York*
  Research Assistant-NuSTAR hard X-ray sources in the Galactic center: HMXB candidates Host: Prof. Charles J. Hailey

Research Interests & Expertise

- **Extremely metal-poor galaxies:** High-energy emission and massive-star feedback at very low metallicity, with implications for early galaxy evolution.
- **Massive stars and stellar winds:** X-ray emission, colliding-wind binaries, and stellar-atmosphere and wind modeling with PoWR.
- **X-ray source populations:** Compact binaries and accreting sources in the Milky Way, globular clusters, and nearby galaxies.
- **Time-domain astrophysics:** X-ray variability, periodicity searches, and AGN quasi-periodic oscillations and eruptions.
- **Observational expertise:** X-ray spectroscopy, timing, and population studies with Chandra, XMM-Newton, NuSTAR, and eROSITA.

Proposals and Awards

- **NASA ADAP, 25-ADAP25-0231; Funded Research Grant** *2026*
  PI: Kaya Mori. Co-Is: **Tong Bao**, *et al.*
- **NuSTAR, Cycle 10; General Observing** *2024*
  PI: Shifra Mandel. Co-Is: **Tong Bao**, *et al*
- **XMM-Newton, Cycle 24; General Observing** *2024*
  PI: Samaresh Mondal. Co-Is: **Tong Bao**, *et al*
- **XMM-Newton, Cycle 24; General Observing** *2024*
  PI: Heshou Zhang. Co-Is: **Tong Bao**, *et al*
- **President's Exceptional Scholarship** *Nanjing university, 2019*
  Annually awarded to the top 10% doctoral student.

- **Scholarship of National Astronomical Observatories** *Chinese Academy of Sciences, 2017*
  Awarded to undergraduate students with strong research experience.
- **People's Scholarship** *Nanjing university, 2017*
  Second class, annually awarded to the top 10% undergraduate students.
- **People's Scholarship** *Nanjing university, 2016*
  Third class, annually awarded to the top 20% undergraduate students.

Selected Oral Presentations

- **X-ray Universe 2026** *Elche, Spain*
  X-ray-luminous late-type giants: an overlooked population contribute to Galactic ridge iron line emission 2026
- **NewAthena Workshop SWG4 2026** *Barcelona, Spain*
  A comprehensive census of X-ray periodic sources in dense star cluster: from Chandra to NewAthena 2026
- **Invited talk, INAF/IASF-MI** *Milan, Italy*
  Periodic X-ray sources: diagnosing the nature and evolution for exotic binaries in dense environment 2025
- **Como Lake center for AstroPhysics Inaugural Workshop** *Como, Italy*
  Unveiling the Soft X-ray Source Population of the Inner Galactic Disk with XMM-Newton 2025
- **Invited talk, Department of Physics, Columbia University** *New York, USA*
  Periodic X-ray sources: diagnosing the nature and evolution for exotic binaries in dense environment 2023
- **eROSITA Scientific Exploitation Workshop** *Suzhou, China*
  Exploring the Outskirts of Globular Clusters with eROSITA 2023
- **Research on the Variations of Active Galactic Nuclei** *Xiamen, China*
  Searching for Quasi-periodic Oscillations in Active Galactic Nuclei of the Chandra Deep Field South 2023
- **Structure, Formation, and Evolution of Star Clusters** *Zhuhai, China*
  Diagnosing dynamical formation and evolution of cataclysmic variables in globular clusters 2023
- **The 60th Anniversary of X-Ray Astronomy** *Beijing, China*
  Searching for Quasi-periodic Oscillations in Active Galactic Nuclei of the Chandra Deep Field South 2022
- **2021-2022 Academic Annual Meeting of Jiangsu Astronomical Society** *Nanjing, China*
  Periodic X-ray Sources in the Galactic Center/Bulge – Application of the Gregory-Loredo Algorithm 2022
- **Chandra Data Science: Novel Methods in Computing and Statistics for X-ray Astronomy** *Online*
  Searching for Quasi-periodic Oscillations of Active Galactic Nuclei in the Chandra Deep Field South 2021
- **The 4th X-ray binary Multi-Band research Symposium** *Kunming, China*
  Diagnosing dynamical origin of exotic binaries in the massive globular cluster 47 Tucanae 2021
- **Chandra Frontiers in Time-Domain Science** *Online*
  Periodic X-ray Sources in the Galactic Center/Bulge – Application of the Gregory-Loredo Algorithm 2020
- **The 3rd X-ray binary Multi-Band research Symposium** *Jinan, China*
  Periodic X-ray Sources in dense environment 2020

Teaching and Supervision

- **Teaching Assistant** *Nanjing university*
  Course: "An Introduction to X-ray Astronomy" 2020, 2021, 2022
  – X-ray data processing training project (https://github.com/baotong6/xray_astronomy), teaching students how to reduce the X-ray data, analyze the spectrum and light curves of CVs, X-ray binaries, ULXs, AGNs, etc.

- **Student Research Supervision**
  2021, 2022, 2025
  – Teodora Darie, **Erasmus traineeships**: "The X-ray spectral properties of different types of main-sequence stars", INAF-OAB
  – Jiachang Zhang, Undergraduate thesis: "Searching for periodic X-ray bianries in the M31 bulge"
  – Jiachang Zhang & Yufan Zhou, Research projects during semesters, "Detecting Emission Line Point Sources in M31 bulge"

Publications (Full List on ADS)

[1] **Bao, T*.**, Ponti, G., et al., 2026, A&A, 710, A243, "X-ray-luminous late-type giants: An overlooked population contributing to the Galactic ridge iron line emission"

[2] **Bao, T*.**, Ponti, G., et al., 2025, A&A, 704, A107, "Unveiling the soft X-ray source population toward the inner Galactic disk with XMM-Newton"

[3] **Bao, T*.**, Li, Z.*, Cheng, Z., Belloni, D. 2024, MNRAS, 527, 7173, "A Chandra Survey of Milky Way Globular Clusters – IV. Periodic X-ray sources"

[4] **Bao, T*.**, Li, Z.*, & Cheng, Z. 2023, MNRAS, 521, 4257, "Periodic X-ray sources in the massive globular cluster 47 Tucanae: Evidence for dynamically formed cataclysmic variables"

[5] **Bao, T*.**, & Li, Z.* 2022, MNRAS, 509, 3504, "Searching for quasi-periodic oscillations in active galactic nuclei of the Chandra Deep Field South"

[6] **Bao, T*.**, & Li, Z.* 2020, MNRAS, 498, 3513, "Periodic X-ray sources in the Galactic bulge: application of the Gregory-Loredo algorithm"

[7] Zhang. J, **Bao. T*.**, Li, Z., 2024, MNRAS, 530, 2096. "A Chandra Search for Periodic X-ray Sources in the Bulge of M31".

[8] Wang Z., Rea N, **Bao, T.**, et al., 2025, Nature, 642, 583, Detection of X-ray emission from a bright long-period radio transient.

[9] Mondal, S., Ponti, G., **Bao, T.**, et al., 2025, A&A, 698, 206. "A sample of ionised Fe line-emitting X-ray sources in the inner Galactic disc".

[10] Mondal, S., Ponti, G., **Bao, T.**, et al., 2024, A&A, 686, 125. "Periodicity from X-ray sources within the inner Galactic disk".

[11] Zhang. J, Li, Z., Hua, Z., **Bao. T.**, 2026, MNRAS, 546, 2237. "Probing periodic and aperiodic variability of X-ray sources in M31, M81, and Centaurus A with Chandra".

[12] Mondal, S., Ponti, G., Filor, L., **Bao, T.**, et al., 2024, A&A, 689, A172, "XMM-Newton and NuSTAR discovery of a likely IP candidate XMMU J173029.8-330920 in the Galactic Disk".

[13] Ma. Q, Zhang. J, Gu. W, Li, Z, Weng. S, **Bao, T.**, 2025, The Innovation, 101181, "The first extragalactic ultra-compact X-ray binary".

Academic Service & Public Outreach

- **PoWR Course, University of Potsdam** — *Potsdam, Germany*
  Local Organizing Committee (LOC) Member — 2026
- **YOUNG, Orienta il tuo futuro** — *Como*
  Volunteer — 2025
- **"Astronomical Night for the Public" Activity of Zuo Dijiang Observatory** — *Nanjing*
  Volunteer — 2021, 2022, 2023
- **Summer Camp for High School Students at the School of Astronomy and Space Science** — *Nanjing*
  Volunteer — 2022
- **Undergraduate Summer Camp at the School of Astronomy and Space Science** — *Nanjing*
  Volunteer — 2021
- **Zuo Dijiang Observatory Open Day Activities** — *Nanjing*
  Lecture Speaker — 2020